EXPERIMENT - 5

Aim:-

Create SQL views for department salary summary and employee hierarchy. Test updatability of views. Implement a recursive CTE to display reporting chains.

Source Code:-

Adding Manager Column

```
ALTER TABLE Employee
ADD manager_id INT;

ALTER TABLE Employee
ADD FOREIGN KEY (manager_id) REFERENCES Employee(emp_id);
```

Setting Managers

```
UPDATE Employee SET manager_id = 101 WHERE emp_id IN (103,108,113,117,127);
UPDATE Employee SET manager_id = 102 WHERE emp_id IN (107,111,116,121,126);
UPDATE Employee SET manager_id = 104 WHERE emp_id IN (110,114,120,125,128);
UPDATE Employee SET manager_id = 105 WHERE emp_id IN (109,115,119,123,130);
UPDATE Employee SET manager_id = 106 WHERE emp_id IN (112,118,124,129);
```

Department Salary Summary View

```
CREATE VIEW Department_Salary_Summary AS
SELECT
   d.dept_id,
   d.dept_name,
   COUNT(e.emp_id) AS employee_count,
   SUM(e.salary) AS total_salary,
   ROUND(AVG(e.salary), 2) AS average_salary
FROM Department d
LEFT JOIN Employee e
   ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name;
```

```
SELECT *
FROM Department_Salary_Summary;
```

Employee Hierarchy View

```
CREATE VIEW Employee_Hierarchy AS
SELECT
   e.emp_id,
   e.emp_name,
   e.manager_id,
   m.emp_name AS manager_name
FROM Employee e
LEFT JOIN Employee m
   ON e.manager_id = m.emp_id;

SELECT *
FROM Employee_Hierarchy;
```

Test Updatability of View

Creating Simple View

```
CREATE VIEW Employee_Salary_View AS
SELECT
   emp_id,
   emp_name,
   salary
FROM Employee;
```

Update View

```
UPDATE Employee_Salary_View
SET salary = 68000
WHERE emp_id = 101;
```

Check Update

```
SELECT *
FROM Employee_Salary_View
WHERE emp_id = 101;
```

Test Department Salary Summary View

```
UPDATE Department_Salary_Summary
SET average_salary = 70000
WHERE dept_id = 1;
```

Recursive CTE - Reporting Chain

```
WITH RECURSIVE ReportingChain AS
(
   SELECT
      emp_id,
      emp_name,
      manager_id,
      0 AS level
   FROM Employee
   WHERE manager_id IS NULL

   UNION ALL

   SELECT
      e.emp_id,
      e.emp_name,
      e.manager_id,
      r.level + 1
   FROM Employee e
   JOIN ReportingChain r
      ON e.manager_id = r.emp_id
)
SELECT
   emp_id,
   emp_name,
   manager_id,
   level
FROM ReportingChain
ORDER BY level, emp_id;
```

Output:-

DEPARTMENT SALARY SUMMARY VIEW

dept_id	dept_name	employee_count	total_salary	average_salary
1	Computer Science Engineering	7	439000.00	62714.29
2	Sciences	6	503000.00	83833.33
3	Law	6	430000.00	71666.67
4	Finance	6	435000.00	72500.00
5	Marketing	5	312000.00	62400.00

EMPLOYEE HIERARCHY VIEW

emp_id	emp_name	manager_id	manager_name
1	Aarav Sharma	null	null
2	Ananya Singh	null	null
3	Rohan Verma	null	null
4	Priya Mehta	null	null
5	Aditya Kumar	null	null
6	Sneha Gupta	null	null
7	Rahul Joshi	null	null
8	Isha Patel	null	null
9	Karan Malhotra	null	null
10	Neha Kapoor	null	null
11	Vikram Rao	null	null

TEST UPDATABILITY OF VIEW

```
Error: The target table Department_Salary_Summary of the UPDATE is not updatable
```

RECURSIVE CTE - REPORTING CHAIN

emp_id	emp_name	manager_id	level
1	Aarav Sharma	null	0
2	Ananya Singh	null	0
3	Rohan Verma	null	0
4	Priya Mehta	null	0
5	Aditya Kumar	null	0
6	Sneha Gupta	null	0
7	Rahul Joshi	null	0
8	Isha Patel	null	0
9	Karan Malhotra	null	0